

An Integrated Computational Platform for Location Specific Design Optimization of Bifacial PV Systems

Pratyai Chakrabarty^{1*}, Aman¹, Ashish Upadhyay¹, Aditya Garg¹, Astitva Kumar¹

¹Department of Electrical Engineering, Netaji Subhas University of Technology, New Delhi, India

*pratyai.chakrabarty.ug22@nsut.ac.in

Abstract—Solar photovoltaic (PV) technology is recognized as a foundational element in the renewable energy transition. Rapid expansion in solar infrastructure is currently being observed. This growth is driven by escalating energy demands. Consequently, available solar irradiance must be captured efficiently so that power generation can be optimized. Significant industry attention is directed toward bifacial photovoltaic modules. Incident sunlight is absorbed by both surfaces of these architectures. Thus, an enhanced energy yield is achieved relative to standard monofacial designs. However, system performance is heavily dictated by installation parameters. The ratio of reflected and direct irradiance received by the rear module face is altered by variables like tilt angle, ground albedo, and mounting height. A computational framework is proposed in this study so the optimal spatial configuration for localized installations can be established. Geographic and temporal inputs are supplied by the operator into a custom platform. Subsequently, hourly meteorological data are extracted from the NASA database. Comprehensive parametric analyses are then executed. The resulting aggregate energy output is quantified so the most efficient configuration can be isolated. An average rear gain of 16.03 percent was recorded for a simulated concrete surface. A standard albedo coefficient of 0.3 was assigned. This gain was reached when tilt angles between 30 and 40 degrees and mounting heights between 350 and 450 cm were implemented. It is demonstrated that maximum total energy output is yielded when elevated mounting heights are paired with moderate tilt angles. Conversely, peak rear irradiance gain is favored when minimal tilt angles are selected.

Keywords—Solar Photovoltaic, Bifacial PV, parametric analysis, computational platform, strategic installation.

I. INTRODUCTION

Solar PV technology is known to be a really promising renewable energy solution. The increasing global demand for clean electricity production is effectively tackled by this technology. Furthermore, the widespread implementation of solar power systems for a variety of climatic regions has been made possible by constant improvements in the design of the modules. Traditionally, electricity is generated by standard monofacial modules using only the sunlight that hits the face panel. Consequently, a large share in available environmental solar energy is left unutilized. Reflected and scattered radiation are included in this wasted energy. To overcome this limitation, bifacial photovoltaic modules have been introduced as a better alternative. Irradiance striking the front and the back surfaces is being successfully converted by all of these advanced systems.

Thus, the maximum total energy is collected from the solar resource [1], [2]. Several studies have shown that bifacial PV systems can have greater energy yields than conventional modules for the appropriate installation configurations are adopted [3]. Detailed reviews of bifacial PV technology emphasize the importance of rear-side irradiance contribution being strongly affected by environmental and geometrical parameters including ground reflectivity, module tilt angle and installation height [1], [4].

Performance of bifacial photovoltaic systems is dependent on the complex interaction between solar radiation, ground reflection, geometry of systems. Numerous studies have studied the effect of these parameters on the overall bifacial PV energy generation modules. For instance, tilt angle has been found to be a critical parameter affecting the balance between the front side and rear side of the irradiance capture, with optimum angles depending on geographical location and solar path characteristics [5], [6]. As such, ground surface reflectivity; often called albedo, plays an important role in the amount of radiation bounce back towards back surface of the module [7], [8]. Experimental investigations and modeling studies have shown that surfaces with more albedo can greatly increase rear-side irradiance and so improve overall energy generation [9], [10]. Besides tilt and albedo, elevation of modules above the ground also has a bearing on the view factor between the module and reflective surfaces, that can affect the amount of the radiation received by the rear side of the panel [11]. Field experiments conducted at various climatic conditions further check environmental factors and installation configurations determine the bifacial gain achievable in real PV systems [12], [13].

Despite the growing body of research about bifacial photovoltaic systems, a number of challenges remain in determining the ideal arrangement of these systems under different environmental conditions. Many existing studies have been focused on individual parameters, such as the tilt angle, albedo or module elevation most often evaluate their effects separately, rather than the sum of their influence on system performance [5], [8], [9]. Other investigations have been developed modeling approaches or global performance assessments of bifacial PV systems; however, these studies often are limited to preset system configurations or certain climatic conditions [4]. Furthermore, studies related to the design of large scale bifacial solar farms show that several

parameters such as orientation of the modules, height of module mounting and system geometry need to be optimized at once to get the maximum energy output [14]. Consequently, there is a need for a systematic analytical framework to be able to evaluate the combined effect of key installation parameters using location specific solar irradiance data. Such an approach would enable the identification of optimal configurations that maximize energy generation for a given site. The present work addresses this requirement by developing a parametric analysis framework that investigates the influence of tilt angle, ground albedo, and mounting height on bifacial PV system performance in order to determine the optimal configuration for maximizing power output.

This paper is divided into four sections – first is the introduction, second is the methodology, third are the results and discussion and fourth is the conclusion.

II. METHODOLOGY

This paper proposes a parametric analysis for the optimal configuration of bifacial photovoltaic systems. Tilt angle, mounting height of the panel, surface albedo and view factor are the parameters that are addressed as shown in Fig 1. A Simulink model was developed initially to model the bifacial PV and later was incorporated in the user frontend to calculate the max power gain achieved and find the best configuration for PV system designers.

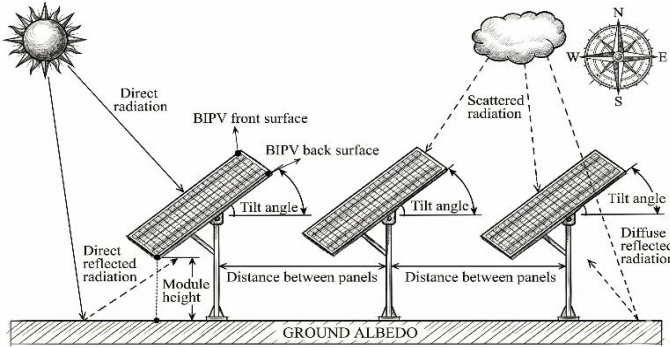


Fig. 1. Schematic representation of parameters required for Bifacial PV system modelling.

A. Mathematical Formulation of Bifacial PV Operation

Unlike conventional monofacial PV modules, which receive irradiance only on the front surface, bifacial modules are capable of utilizing additional radiation reflected from the ground and surrounding surfaces. The total effective irradiance on a bifacial module can therefore be expressed as the sum of the front and rear irradiance components [4]:

$$G_{tot} = G_{front} + G_{rear} \quad (1)$$

where; G_{tot} = total irradiance received by the bifacial module, G_{front} = irradiance incident on the front surface, G_{rear} = irradiance incident on the rear surface. The front irradiance and rear irradiance depend on several factors which we take into account in the analysis. The front-side irradiance is estimated using the Liu–Jordan isotropic sky model [15], which

decomposes the global horizontal irradiance into beam, diffuse, and ground-reflected components. The direct beam irradiance incident on the tilted module surface is calculated as:

$$B_t = DNI \times \max(0, \cos I) \quad (2)$$

where; DNI represents the direct normal irradiance and I denotes the angle of incidence between the incoming solar radiation and the module surface. The $\max(0, \cos I)$ term ensures that negative irradiance values are avoided when the sun is below the panel horizon. The diffuse irradiance component is modeled assuming isotropic sky distribution. The sky view factor and ground view factor for the tilted surface is given by [15]:

$$sky_{vf} = \left(\frac{1 + \cos \beta}{2} \right) \quad (3)$$

$$gnd_{vf} = \left(\frac{1 - \cos \beta}{2} \right) \quad (4)$$

where β represents the module tilt angle. The sky view factor quantifies the portion of the sky dome visible to the tilted PV surface. Similarly, the ground view factor represents the fraction of ground-reflected radiation reaching the module surface. The diffuse component of global horizontal irradiance (GHI) is given by DHI (direct horizontal irradiance). Combining these contributions, the total front-side irradiance is calculated as [15]:

$$G_{front} = B_t + (DHI \times sky_{vf}) + (GHI \times \rho \times gnd_{vf}) \quad (5)$$

The rear irradiance to be computed depends on shadow view factor F_v , albedo, DHI, GHI and rear sky view factor. The shadow view factor F_v represents the reduction in rear irradiance due to row-to-row shading in the PV array. It is determined by evaluating the solar profile angle and the geometric relationship between the sun position and the module orientation. The solar profile angle is computed as [16], [17]:

$$\gamma_{profile} = \tan^{-1} \left(\frac{\tan(sun_{elev})}{\cos(sun_{az} - panel_{az})} \right) \quad (6)$$

where; sun_{elev} is the solar elevation angle and sun_{az} and $panel_{az}$ represent the solar and panel azimuth angles, respectively. The lower and upper bounds of the shadow region are calculated using (6) and is given by [16], [17]:

$$shd_l = \frac{h}{\tan(\gamma_{profile})} \quad (7)$$

$$shd_u = \frac{h + W \sin \beta}{\tan(\gamma_{profile})} \quad (8)$$

where; h is the mounting height and W is the width of the module. The rear sky view factor ($rear_{vf}$) geometrically is

represented by (3). The shadow view factor F_V is now calculated as [18]:

$$F_V = rear_{vf} - \int_{shd} \frac{\cos\theta_1 \cos\theta_2}{2r} dx \quad (9)$$

where; θ_1 and θ_2 represent the geometric angles between the ground and module surfaces and r is the distance between the differential elements as shown in Fig 2. Now, the final rear irradiance is given by incorporating the above factors [18]:

$$G_{r0} = \rho \cdot DHI \cdot rear_{vf} + \rho \cdot (GHI - DHI) \cdot (rear_{vf} - F_V) \quad (10)$$

$$G_{rear} = G_{r0} \times bifaciality \quad (11)$$

The bifaciality factor represents the ratio of rear-side efficiency to front-side efficiency of the bifacial PV module.

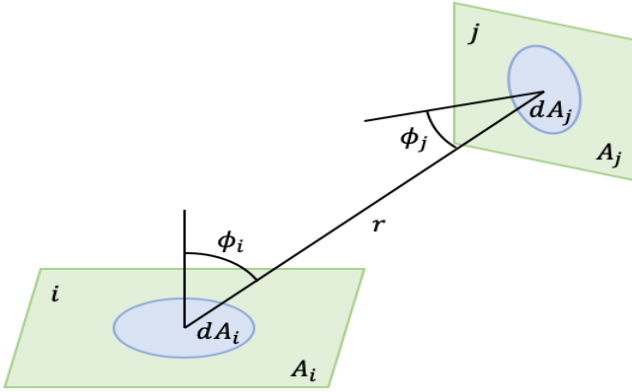


Fig. 2. Geometric view factor between differential ground and module surface elements for rear irradiance computation.

B. Simulink Model

The bifacial PV system is modeled in MATLAB/Simulink as shown in Fig. 3 to analyze the electrical performance of the module under varying environmental and installation conditions. The model incorporates key parameters such as GHI, ground reflectivity (albedo), tilt angle, mounting height, and temperature.

The input parameters are processed through a MATLAB function block (*calculate_irradiance*), which determines the effective irradiance G_{eff} received by the module, the one as in (1). The resulting effective irradiance value is then supplied to the PV Array block as the irradiance input I_r . The PV Array block represents the electrical model of the photovoltaic module and produces the output current and voltage corresponding to the applied operating conditions. To obtain the electrical characteristics of the PV system, a controlled voltage source is used to perform an I-V sweep across the PV module terminals. The voltage sweep is generated using a ramp signal (V_{ramp}), which gradually varies the applied voltage across the PV array over the simulation interval. As the voltage changes, the PV module produces corresponding current values, allowing the complete I-V characteristic curve of the

module to be obtained. The measured current (I_{meas}) and voltage (V_{meas}) signals are collected through a bus creator block and forwarded to subsequent processing stages. The first scope plots the current-voltage (I-V) characteristics of the bifacial PV module, while the second scope plots the power-voltage (P-V) characteristics.

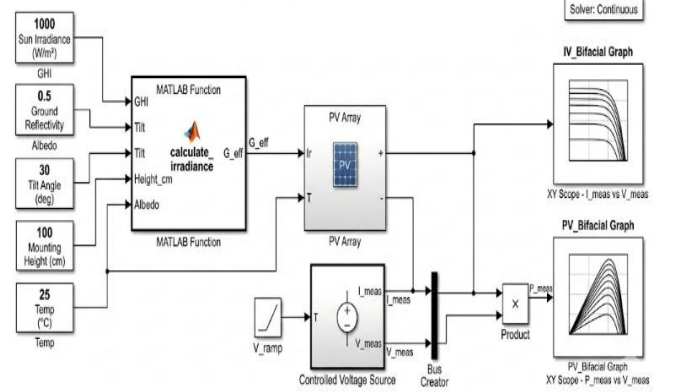


Fig. 3. Simulink model for Bifacial Power Computation

C. Proposed Platform Framework

The proposed framework is designed to evaluate and determine the optimal configuration of bifacial photovoltaic (PV) systems by integrating location-based solar irradiance data with parametric performance analysis as shown in Fig.4. The framework consists of four primary layers: the user interface, backend application, analysis and simulation engine, and the results dashboard.

The process begins at the user interface, where the user provides the necessary input parameters required for the analysis. These inputs include the geographical location in the form of a selected city, the desired date range for analysis, and key system configuration parameters such as module tilt angle, mounting height, and ground surface type. Once the input parameters are specified, then, these inputs are processed by the backend application. After that a data request for solar irradiance is generated by the system. The required irradiance information is then collected using the NASA Power API. Specific solar radiation information on the selected city and time period is made available directly via this programming interface.

After that, the irradiance dataset is sent to the analysis and simulation engine. The computation analysis of the bifacial photovoltaic system is then taken care of by this particular module. The data collected from irradiance reading and the customized installation parameters are synthesized together by the engine. As a result the electrical behaviour of the bifacial setup is successfully simulated. The current versus voltage characteristics, and the power versus voltage characteristics of the system are tested during this process. After this, various configuration settings for key performance indicators such as total output power and bifacial gain are calculated.

Finally, the computed results are presented via a results dashboard, this offers a structured visualization of the system performance. The dashboard displays the maximum power

output and bifacial gain values in tabular form, along with graphical representations of the I–V and P–V characteristics.

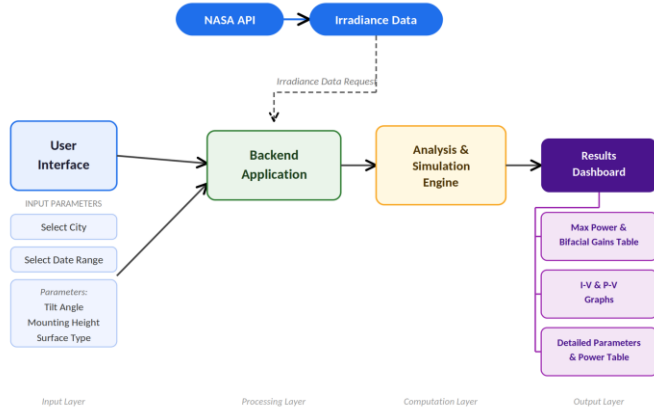


Fig. 4. Four-layer framework integrating NASA irradiance data for parametric bifacial PV optimization.

III. RESULTS AND DISCUSSION

To evaluate the proposed framework, a case study was conducted for the location Greater Noida, India, using the developed frontend–backend analysis platform. The geographical location was selected through the user interface, and the simulation date was fixed as 7 March 2026 in order to obtain location-specific solar irradiance data for that day. a parametric sweep was performed over a range of geometric configurations. The module mounting height was varied from 50 cm to 450 cm with increments of 50 cm, while the tilt angle was varied from 10° to 50° in steps of 10°. The azimuth angle of the PV module was fixed at 180° for south facing panels, which is commonly used for photovoltaic installations in the northern hemisphere to maximize solar exposure. The bifaciality factor is assumed practically to be 0.7. The ground surface type was assumed to be concrete, corresponding to an albedo value of 0.3. The Nominal Operating Cell Temperature (NOCT) is kept at 45°C. The irradiance was fetched using the NASA Power API and consequently, the observed total GHI was 6.08 kWh/m² and peak GHI was 866.53 Wh/m².

The analyses were run using MATLAB engine incorporating the mathematical formulation discussed before. The optimal configuration among 45 evaluation points was available as a result in form of optimal tilt angle (30°) and mounting height (450 cm) corresponding to the selected albedo. The rear gain came out to be a good 16.55% using (11). The Fig 5. shows the I–V and P–V graphs for the various parameter based analyses. The optimization objective can be max energy or max rear gain based on which respective configuration can be achieved. The PV system designer can choose what is his optimization objective. Table I lays down the practical albedo values for different surface types. Table II shows the rear gain percentage, energy and power achieved for the three parameters with optimization objective of max energy. Table III shows the same but for optimization objective of max rear gain.

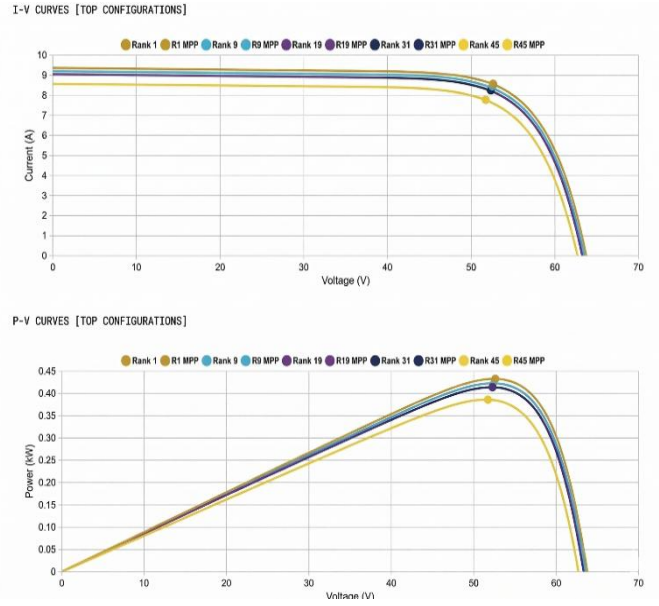


Fig. 5. I–V and P–V characteristics for different configurations

The sweep parameters were varied all together in the first analyses. There is also a provision in the frontend wherein you can fix two parameters and vary the third to get another set of optimal configurations. This makes the platform dynamic and according to the practical needs of the PV system designers. Fig. 6 shows the variation of rear share of effective irradiance on varying albedo (surface type) with fixed parameters, tilt angle of 20°, height of 100 cm, azimuthal angle of 180° and bifaciality of 0.7. Consequently, we achieved a rear gain share of 24.72% for aluminium surface. Fig. 7 shows the variation of rear share of effective irradiance on varying panel height with tilt angle fixed at 20°, albedo of 0.18 and rest all factors remain same as in case of Fig 6. With this variation, we achieve 6.61% for a panel height of 100 cm. Fig. 8 shows the variation of rear share of effective irradiance on varying tilt angle with height fixed at 100 cm, albedo of 0.18 and rest all factors remain same as in case of Fig.6. As a result, we obtain a maximum rear gain share of 7.24% for tilt angle of 10°.

TABLE I. ALBEDO VALUES OF COMMON SURFACE TYPES [11]

Surface Type	Albedo Value
Urban environment	0.14-0.22
Grass	0.15-0.25
Fresh Snow	0.82
Concrete	0.25-0.35
Aluminum	0.85
Wet Asphalt	0.18
Red Tiles	0.33

TABLE II. MAX ENERGY BASED OPTIMAL CONFIGURATION

Rank	Height (cm)	Tilt angle (°)	Surface type	Energy (kWh)	Peak power (kW)	Rear gain (%)
1	450	30	Concrete	3.16843	0.43249	16.5518
2	450	40	Concrete	3.16570	0.43154	15.6057
3	400	30	Concrete	3.16521	0.43201	16.3917
4	400	40	Concrete	3.16258	0.43108	15.4506
5	350	30	Concrete	3.16113	0.43140	16.1891

TABLE III. MAX REAR GAIN BASED OPTIMAL CONFIGURATION

Rank	Height (cm)	Tilt angle (°)	Surface type	Energy (kWh)	Peak power (kW)	Rear gain (%)
1	450	10	Concrete	2.99422	0.41002	17.8989
2	400	10	Concrete	2.99130	0.40957	17.7561
3	350	10	Concrete	2.98755	0.40899	17.5731
4	300	10	Concrete	2.98258	0.40823	17.3301
5	450	20	Concrete	3.11109	0.42532	17.3252

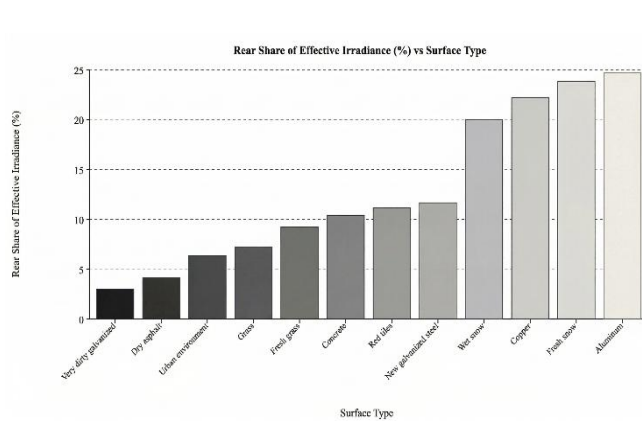


Fig. 6. Rear share of effective irradiance on varying albedo

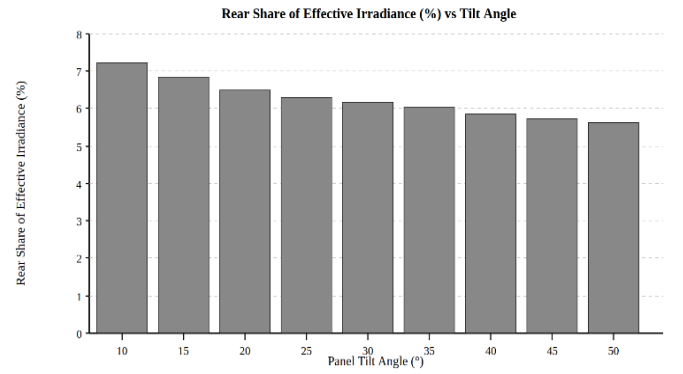


Fig. 8. Rear share of effective irradiance on varying Tilt Angle

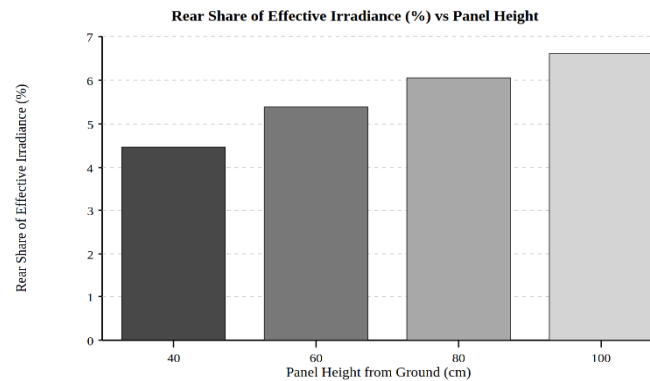


Fig. 7. Rear share of effective irradiance on varying height

IV. CONCLUSION

This paper analyses the optimal configuration of bifacial PV systems through a computational framework that integrates location-specific irradiance data with parametric performance evaluation. The developed platform uses irradiance data obtained from the database of the National Aeronautics and Space Administration Power program, and processed via the simulation engine written in the programming language of MATLAB, implementing the models of irradiance and electricity of the PV modules of bifacial type. The results show that the contribution of the irradiance at the backside is strongly affected by the interaction of the surface reflecting ability, the tilt angle of the modules and the mounting height. The higher ground albedo will result in more rear irradiance via greater reflection. In contrast, variances in mounting height against other parameters that were fixed cause relatively smaller variances in rear gain, since the geometric view factor between the ground and the module does not increase substantially past

certain elevations. Similarly, changes in the tilt angle alter the balance of sky and ground view factors which in turn influences the proportion of reflected irradiance that reaches the rear side. We get average rear gain percentage of 16.03 for concrete surface type with tilt angle in the variable range of 30°-40° and height of 350 cm – 450 cm. These conclusions prove that bifacial PV performance is highly sensitive to installation geometry and surface characteristics around solar panels and the formulated framework offers a systematic and flexible solution for determining the ideal system configurations based on various design purposes.

V. REFERENCES

- [1] R. Guerrero-Lemus, F. Vega, E. Kim, A. Kimm and M. Shephard, "Bifacial solar photovoltaics – A technology review," *Renewable and Sustainable Energy Reviews*, vol. 60, pp. 1533-1549, 2016.
- [2] M. Alam, M. R. Khan and C. Deline, "Performance comparison between bifacial and monofacial photovoltaic systems under varying environmental conditions," *Sustainable Energy Technologies and Assessments*, vol. 57, p. 103210, 2023.
- [3] C. Deline, A. A. Pelaez, S. MacAlpine and B. Marion, "Bifacial photovoltaic system performance: Separating fact from fiction," National Renewable Energy Laboratory (NREL), Golden, CO, USA, 2019.
- [4] X. Sun, M. R. Khan, C. Deline and M. A. Alam, "Optimization and performance of bifacial solar modules: A global perspective," *Applied Energy*, vol. 212, pp. 1601-1610, 2018.
- [5] A. Basak, S. Das and R. Banerjee, "Tilt angle optimization for bifacial photovoltaic modules for enhanced energy generation," *Applied Energy*, vol. 345, p. 121245, 2025.
- [6] U. Peter and M. Novak, "Optimal tilt angle for maximizing energy production of bifacial solar panels," *Journal of Sustainable Energy Systems*, vol. 14, no. 1, pp. 45-56, 2025.
- [7] N. Riedel-Lyngskær, J. M. L. P. Larsen and B. G. Nielsen, "Effect of spectral albedo on energy yield of bifacial photovoltaic systems," *Solar Energy*, vol. 231, pp. 176-186, 2022.
- [8] M. H. Aksoy and H. A. Ceylan, "Investigation of ground albedo effects on bifacial photovoltaic panel performance," *Renewable Energy Research Journal*, vol. 13, no. 4, pp. 1800-1808, 2023.
- [9] N. Baghel, A. Chandel and S. K. Sharma, "Performance evaluation and optimization of albedo and tilt angle in bifacial photovoltaic systems," *Solar Energy*, vol. 250, pp. 84-95, 2023.
- [10] I. N. Atalay, M. Kaya and H. Ozturk, "Experimental investigation of ground surface reflectivity impact on bifacial photovoltaic systems," *Solar Energy*, vol. 246, pp. 65-74, 2023.
- [11] F. Dincer and E. Ozer, "Optimization of rear-side energy contribution in bifacial PV panels: A parametric analysis on albedo, tilt, height, and mounting configuration," *Energies*, vol. 18, p. 4443, 2025.
- [12] D. S. Braga, R. Zilles and E. Lorenzo, "Performance analysis of bifacial photovoltaic modules under tropical climatic conditions," *Renewable Energy and Environmental Sustainability*, vol. 8, pp. 1-10, 2023.
- [13] A. F. Almarshoud, A. Al-Badi and M. Al-Sulaiman, "Experimental performance analysis of bifacial photovoltaic modules under different operating conditions," *Energies*, vol. 17, no. 21, p. 5456, 2024.
- [14] M. T. Patel, M. R. Khan and M. A. Alam, "Optimum design of tracking bifacial solar farms," *IEEE Journal of Photovoltaics*, vol. 11, no. 1, pp. 237-245, 2021.
- [15] B. Y. H. Liu and R. C. Jordan, "The interrelationship and characteristic distribution of direct, diffuse and total solar radiation," *Solar Energy*, vol. 4, pp. 1-19, 1960.
- [16] J. A. D. Beckman and W. A., *Solar Engineering of Thermal Processes*, Hoboken, NJ, USA: Wiley, 2013.
- [17] B. Marion, C. Deline, A. A. Pelaez, S. MacAlpine and B. Stein, "A practical irradiance model for bifacial PV modules," *IEEE Journal of Photovoltaics*, vol. 7, no. 1, pp. 1-7, 2017.
- [18] U. A. Yusufoglu, T. M. Pletzer, H. H. Kurz and H. J. Hübner, "Analysis of the performance of bifacial modules in vertical and tilted installations," *Progress in Photovoltaics: Research and Applications*, vol. 23, no. 12, pp. 1820-1835, 2015.